All Local Minima Are Global

Laurent & von Brecht (ICML, 2018) | Deep Linear Networks with Arbitrary Loss

Paper in one minute. Can adding many *linear* layers create bad local minima even when the loss on the combined linear map is convex? **Under two assumptions, no: every local minimum in the separate layer weights is globally optimal.** This is a result about *landscape geometry*, not a guarantee that gradient descent trains quickly.

1. First principles: what are the two optimization problems?

With one linear transformation, write A for the combined map and minimize $f(A)$. A deep linear network replaces A by a product $A = W_L W_{L-1} \cdots W_1$, so it minimizes

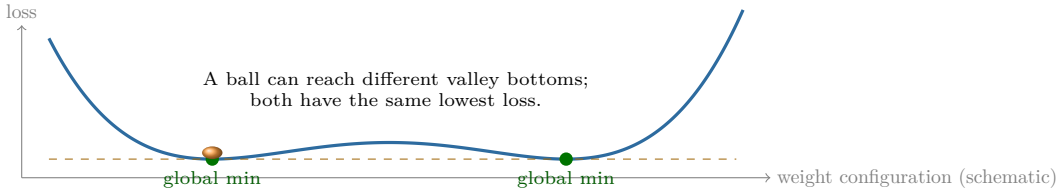
$$\boxed{F(W_1, \ldots, W_L) = f(W_L \cdots W_1).}$$

Why can F be non-convex? Even if f is convex in A, multiplying separately trainable weights makes F nonlinear in those weights. For one input and two scalar layers, with target 4:

$$f(A) = (A-4)^2 \quad \text{(convex parabola)}, \qquad F(w_1, w_2) = (w_2 w_1 - 4)^2 \quad \text{(non-convex)}.$$

A **local minimum** means no *sufficiently small* change of the weights lowers the loss; a **global minimum** means *no change anywhere* can lower it. There may be many different optimal weight settings: $(2, 2)$ and $(1, 4)$ both give $w_2 w_1 = 4$ and zero loss.

2. Visual intuition: many valleys, but none is a bad valley



Important: This sketch is conceptual, *not* a literal cross-section of the paper's high-dimensional surface. Saddles and flat regions can still exist: "no bad local minima" $\neq$ "optimization is easy."

3. The crucial theorem: why can they be sure?

What needs proving? It is *not* enough that each layer gradient vanishes. By the chain rule, $\partial F/\partial w_1 = w_2 f'(A)$ and $\partial F/\partial w_2 = w_1 f'(A)$; at $(0, 0)$ they both vanish although $f'(0) = -8$. That point is stationary but not a local minimum.

The actual bridge established by the paper (Theorem 3): for a *local minimum* $(W_1^*, \ldots, W_L^*)$ of the factorized objective, the combined matrix $A^* = W_L^* \cdots W_1^*$ satisfies $\nabla f(A^*) = 0$, provided the loss is differentiable and the layers are sufficiently wide. Since f is also convex, $\nabla f(A^*) = 0$ implies $f(A^*) \leq f(A)$ for *all* A. Thus the original local minimum is global.

$$\boxed{\text{local minimum of } F \implies \nabla f(A^*) = 0 \overset{f \text{ convex}}{\implies} \text{global minimum.}}$$

4. What assumptions make this true?

(i) The outer loss f is **convex and differentiable**. **(ii)** Every hidden-layer width d_k satisfies $d_k \geq \min(d_0, d_L)$ (the thinnest layer is an input or output layer). This avoids a rank bottleneck that would restrict which combined maps can be represented. The paper constructs a *convex, Lipschitz but nondifferentiable* loss for which a bad local minimum exists—so differentiability really matters.

Limits / survey connection. The theorem does *not* cover arbitrary nonlinear networks, nonconvex outer losses, or narrow bottleneck architectures; nor does it promise fast convergence or escape from saddles. **RQ1:** non-convex parameterizations can have benign minima. **RQ4:** that claim depends strongly on loss smoothness and network width.

Takeaway: Adding linear layers changes the shape of the optimization problem but, under the theorem's assumptions, does not create suboptimal local minima. *Source:* Laurent & von Brecht, "Deep Linear Networks with Arbitrary Loss: All Local Minima Are Global," ICML (PMLR 80), 2018, Theorems 1–3.